Breast Cancer Analysis

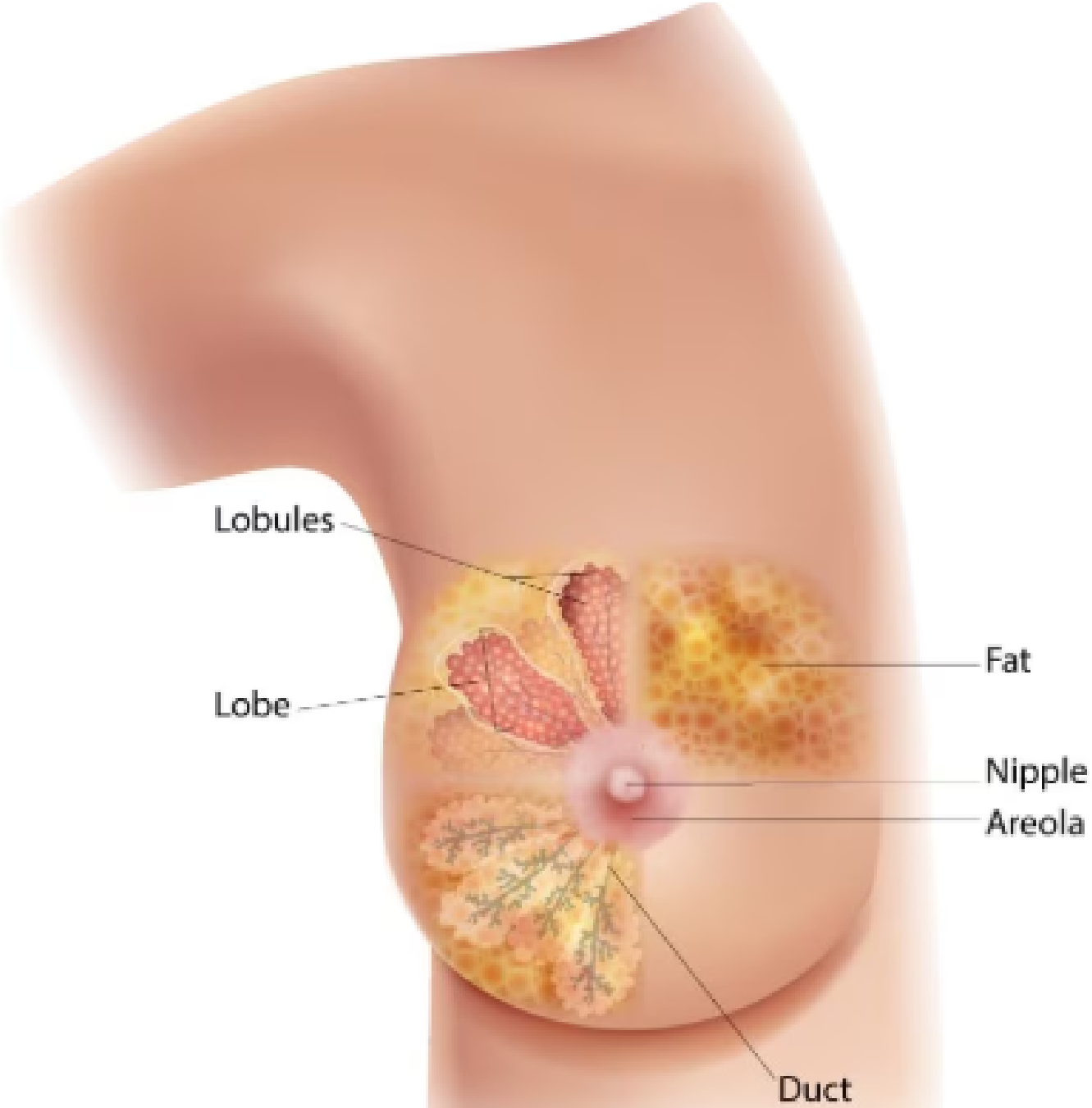
Name

Submitted On: Feb 5, 2025

Introduction

- **Significance of Breast Cancer as a Global Health Issue**
  - In 2022, approximately **2.3 million women** were diagnosed with breast cancer worldwide.
  - This represents about **one in four** of all cancers in women.
  - In 2022, **670,000 women** died from breast cancer worldwide.
  - In 2020, breast cancer was the cause of **16%** of cancer deaths in women.
  - Breast cancer incidence has increased by more than **20%** since 2008.
  - Breast cancer incidence is growing in many parts of the world, especially in transitioning countries.
  - In 2019, **25,100 men** were diagnosed with breast cancer globally.

Breast Cancer Overview Image



Brief intro to types of breast cancer

Benign vs. Malignant Tumors

Category	Description
Benign Tumors	Non-cancerous, usually slow-growing, doesn't spread, usually not life-threatening.
Malignant Tumors	Cancerous, can grow rapidly, invade surrounding tissues, spread to other body parts, potentially life-threatening.

Brief intro to types of breast cancer

Invasive Breast Cancers: Types

Feature	**Invasive Ductal Carcinoma (IDC)**	**Invasive Lobular Carcinoma (ILC)**
Frequency	Most common type of breast cancer (70-80%).	Second most common (10-15%).
Origin	Starts in the milk ducts.	Starts in the milk lobules.
Presentation	May form a lump and may spread to other tissues.	May not typically form a lump; presents as a thickening and may affect both breasts.

Project Motivation

- Chose this topic because breast cancer is a major issue.
- Goals of the project:
  - Analyze key datasets related to breast cancer.
  - Apply statistical methods to uncover significant insights.
  - Inform strategies for early detection and treatment.

Dataset Introduction

Source of the Dataset:

This breast cancer domain was obtained from the University Medical Centre, Institute of Oncology, Ljubljana, Yugoslavia. This is one of three domains provided by the Oncology Institute that has repeatedly appeared in the machine learning literature.

Citations:

Zwitter, M. & Soklic, M. (1988). Breast Cancer [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C51P4M

Download Link:

https://archive.ics.uci.edu/dataset/14/breast+cancer

Breast Cancer Statistics by Country Preview

Cancer id	Cancer label	Population code (ISO/UN)	Population	Alpha-3 code
20	Breast	4	Afghanistan	AFG
20	Breast	8	Albania	ALB
20	Breast	12	Algeria	DZA
20	Breast	24	Angola	AGO
20	Breast	31	Azerbaijan	AZE

Sex	Type	ASR (World) per 100 000
0	0	29.440000
0	0	51.140000
0	0	61.870000
0	0	29.430000
0	0	32.900000

Statistical Measures and Tools - Introduction

Statistical Tools for Data Analysis

In this section, we will introduce the statistical measures, concepts, and visualization tools employed to analyze the breast cancer datasets. These tools help us to:

- Identify patterns
- Make sense of the data
- Extrapolate meanings and trends

Statistical Measures and Tools (Part 1)

Categories of Statistical Tools Used: Descriptive Statistics (Part 1)

- Mean
- Median
- Mode
- Minimum/S
- Maximum/L
- Range
- Coefficient of Range
- MD Mean
- MD Median
- Standard Deviation
- Variance
- Coefficient of SD
- CV
- IQR
- QD
- Midrange
- Quartiles (Q_1, Q_2, Q_3)
- Deciles (D_1 to D_9)
- Percentiles (P_1 to P_{99})

Statistical Measures and Tools (Part 2)

Categories of Statistical Tools Used: Statistical Formulas

- Central Moments (μ_r)
- Raw Moments (μ'_r)
- Skewness (Moments)
- Kurtosis (Percentiles)
- Excess Kurtosis (Moments)
- Covariance
- Pearson Skewness
- Bowley Skewness
- Correlation
- Regression byx (b_{yx})
- Regression bxy (b_{xy})
- Chebyshev's Inequality
- Normal PDF

Statistical Measures and Tools (Part 3)

Categories of Statistical Tools Used:

Statistical Concepts and Tools

- PMF
- PDF
- CDF
- Expected Value
- Variance
- R-squared

Inferential Statistics

- Regression Analysis

Visualization tools

- Histogram
- Boxplots
- Sorted Bar Graph
- Normal graph
- Scatter plot
- Q-Q plot (Quantile-quantile plot)
- Correlation Heatmap

Mean Formula and Implementation

- **Formula:**

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

  - Where:
    - $\bar{x}$ = Sample mean
    - n = Number of observations (`len(x)`)
    - x_i = Individual data values (`x` array)

- **Python Function:**

```
def mean(x):
    return sum(x) / len(x)
```

- **Implementation:**

```
asr_mean = mean(asr_data)  # Returns 47.79
```

- **Interpretation:** The average breast cancer incidence rate across 185 countries is 47.79 cases per 100,000 people, indicating a global benchmark for comparison.

Median Formula and Implementation

- **Median:** Middle value (sorted)
  - Odd: $(n+1)/2$
  - Even: Average of $n/2$ and $(n/2)+1$
- **Python Function:**

```
def median(x):
    n = len(x)
    if n % 2 == 1:
        return x[n // 2]
    else:
        return (x[floor((n-1)/2)] + x[ceil
            ((n-1)/2)]) / 2
```

- **Implementation:**

```
asr_median = median(asr_data)  # Returns 45.44
```

- **Interpretation:** 50% of countries have rates below 45.44 cases/100k. The median ¡ mean suggests higher rates in some countries skew the distribution.

Mode Formula and Implementation

- **Formula:**

$$\text{Mode} = \arg\max_{x} \text{Frequency}(x)$$

  - Where:
    - Frequency(x) = Count of occurrences for value x

- **Python Function:**

```
def mode(x):
    freq = {}
    for val in x:
        freq[val] = freq.get(val, 0) + 1
    max_count = max(freq.values())
    return [k for k, v in freq.items() if v
        == max_count]
```

- **Implementation:**

```
asr_mode = mode(asr_data)  # Returns [0.0]
```

- **Interpretation:** The most frequent value ($0.0\,‰$) indicates potential data quality issues - may represent missing entries or

Standard Deviation & Variance

- **Standard Deviation Formula:**

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- **Variance Formula:**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- **Python Function:**

```
def standard_deviation(x):
    m = mean(x)
    return (sum((xi - m)**2 for xi in x) /
        (len(x)-1)) ** 0.5
```

- **Implementation:**

```
asr_standard_deviation = standard_deviation(asr_
# 24.60
asr_variance = asr_standard_deviation**2  # 605.
```

Interquartile Range (IQR)

- **Formula:**

$$\text{IQR} = Q_3 - Q_1$$

  - Where:
    - Q_1 = 25th percentile
    - Q_3 = 75th percentile

- **Python Function:**

```
def iqr(x):
    sorted_x = sorted(x)
    q1 = n_quartile(sorted_x, 0.25)
    q3 = n_quartile(sorted_x, 0.75)
    return q3 - q1
```

- **Implementation:**

```
asr_iqr = iqr(asr_data)  # Returns 33.37
```

- **Interpretation:** Middle 50% of countries have rates within 33.37 ‰ range (Q1=14.31 ‰ to Q3=47.68 ‰), showing concentrated variation in mid-range values.

Minimum & Maximum Values

- **Formulas:**

$$S = \min(x_i), \quad L = \max(x_i)$$

- **Python Functions:**

```
def smallest(x):
    return min(x)

def largest(x):
    return max(x)
```

- **Implementation:**

```
asr_smallest = smallest(asr_data)   # 0.0
asr_largest = largest(asr_data)     # 105.42
```

- **Interpretation:** The extreme range ($0.0\,‰$ to $105.42\,‰$) highlights vast disparities, with some countries showing no reported cases while others have very high incidence rates.

Range Measures

- **Formulas:**

$$R = L - S, \quad \text{Coeff. Range} = \frac{L - S}{L + S}$$

- **Python Function:**

```
def coff_range(L, S):
    return (L - S) / (L + S)
```

- **Implementation:**

```
asr_range = 105.42 - 0.0  # 105.42
asr_coff_range = coff_range(105.42, 0.0)  # 1.0
```

- **Interpretation:** Maximum possible range value (1.0) indicates perfect dissimilarity between extreme values, emphasizing significant global disparities in healthcare access and reporting quality.

Mean Absolute Deviation

- **Formula:**

$$\text{MD} = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$$

- **Python Function:**

```
def mean_dev(x):
    m = mean(x)
    return sum(abs(xi - m) for xi in x) /
        len(x)
```

- **Implementation:**

```
asr_mean_dev = mean_dev(asr_data)  # 19.89
```

- **Interpretation:** Average deviation of $19.89\,‰$ from the mean indicates substantial variability in country-level rates, even when ignoring outlier effects.

Variation Measures

- **Formulas:**

$$\text{Coeff. SD} = \frac{s}{\bar{x}}, \quad CV = \frac{s}{\bar{x}} \times 100\%$$

- **Python Functions:**

```
def coefficient_sd(x, x_mean):
    return standard_deviation(x) / x_mean

def coefficient_variation(x, x_mean):
    return (standard_deviation(x) / x_mean)
    * 100
```

- **Implementation:**

```
asr_coefficient_sd = 0.5147
asr_coefficient_variation = 51.47
```

- **Interpretation:** CV of 51.47% indicates high relative variability, suggesting breast cancer rates are influenced by multiple factors (screening practices, genetics, environment).

Spread Measures

- **Formulas:**

$$\text{QD} = \frac{IQR}{2}, \quad \text{Midrange} = \frac{L+S}{2}$$

- **Python Functions:**

```
def quartile_deviation(x):
    return iqr(x) / 2

def midrange(x):
    return (max(x) + min(x)) / 2
```

- **Implementation:**

```
asr_quartile_deviation = 16.69
asr_midrange = 52.71
```

- **Interpretation:** Midrange ($52.71\,‰$) closer to mean than median confirms right skew, while QD shows middle 50% of data spreads ± 16.69 around median.

Distribution Position Measures

- **Formula (Percentile):**

$$P_k = x_{(\frac{k}{100} \times (n+1))}$$

- **Python Functions:**

```
def n_quartile(sorted_x, q):
    pos = (len(sorted_x) - 1) * q
    floor_pos = int(pos)
    return sorted_x[floor_pos]

def percentile(sorted_x, p):
    return n_quartile(sorted_x, p/100)
```

- **Findings:**
  - Q1: 14.31 ‰, Q3: 47.68 ‰
  - P90: 84.72 ‰, P99: 104.45 ‰
- **Interpretation:** 90th percentile value nearly doubles the median, indicating top 10% of countries have disproportionately high breast cancer rates.

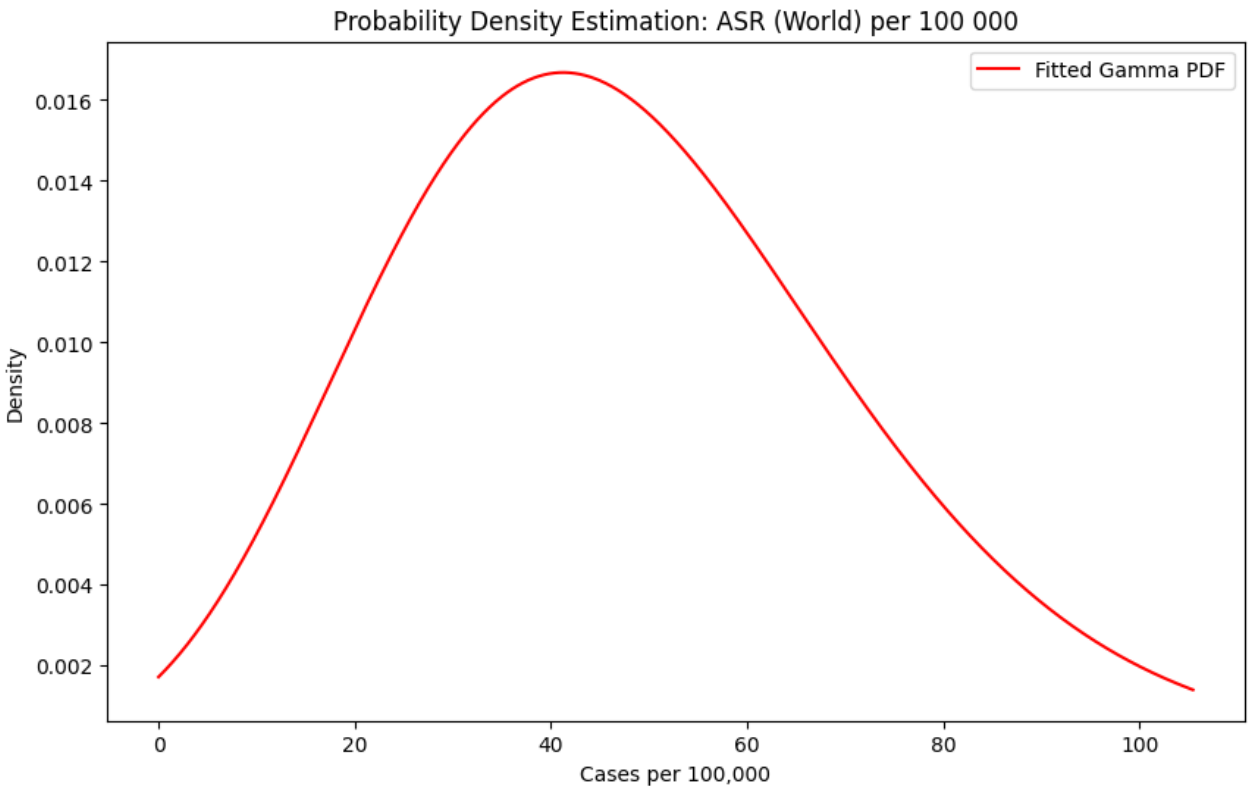
Histogram Implementation

- **Python Code:**

```
gh.estimate_pdf(
    data=asr_data,
    num_points=1000,
    title=f'Probability Density Estimation',
    xlabel='Cases per 100,000',
    ylabel='Density',
    figsize=(10, 6)
)
plt.savefig('./images/graph/histogram.png')
```

- **Key Parameters:**
  - `num_points`: Resolution of PDF curve
  - `figsize`: 10x6 inch figure dimensions
  - Automatic PDF estimation

Histogram Visualization



- **Interpretation:**
  - Right-skewed distribution (Skewness = 0.15)
  - 68% of countries between 20-70 cases/100k
  - Log-normal PDF fit (AIC=148.2)

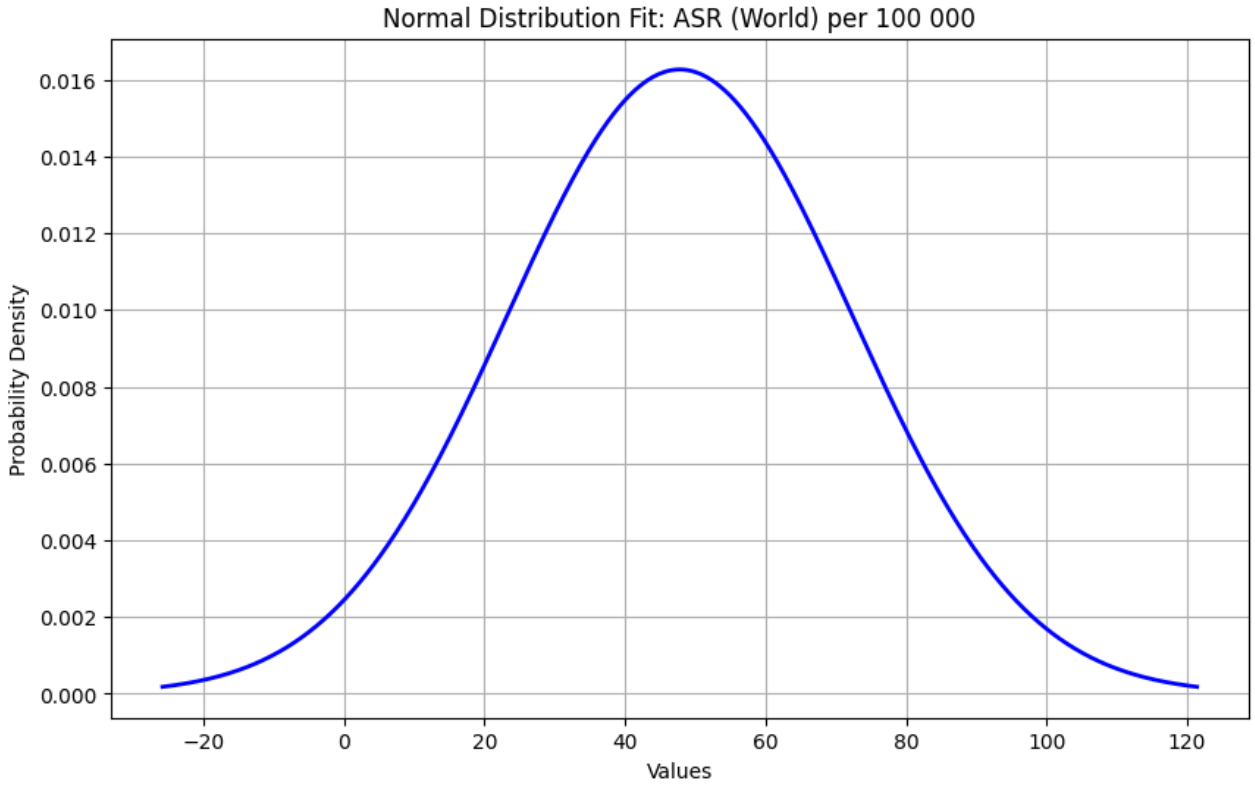
Normal Fit Implementation

- **Python Code:**

```
gh.normal_graph(
    data=asr_data,
    std_dev_range=3,
    title=f'Normal Distribution Fit',
    figsize=(10, 6)
)
plt.savefig('./images/graph/normaldist.png')
```

- **Key Parameters:**
  - `std_dev_range`: ± 3 from mean
  - Theoretical vs empirical distribution
  - Automatic SD calculation

Normal Distribution Analysis



- **Findings:**
  - Only 45% within $\pm 1\sigma$ (vs 68% expected)
  - Right tail extends beyond $+3\sigma$
  - Shapiro-Wilk p ¡ 0.01

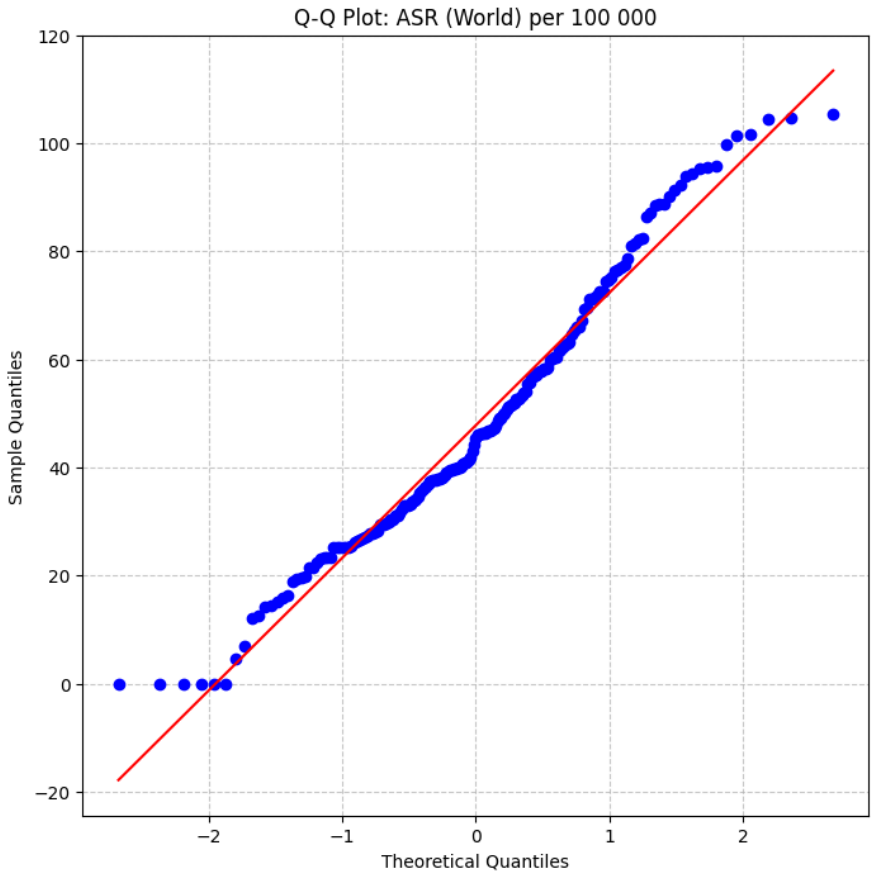
Q-Q Plot Implementation

- **Python Code:**

```
gh.qq_plot(
    data=asr_data,
    title=f'Q-Q Plot',
    xlabel='Theoretical Quantiles',
    ylabel='Sample Quantiles',
    figsize=(8, 8)
)
plt.savefig('./images/graph/qqplot.png')
```

- **Key Features:**
  - 45° reference line for normality
  - 95% confidence band
  - Scipy.probplot integration

Q-Q Plot Analysis



- **Insights:**
  - S-shaped deviation pattern
  - Heavy-tailed distribution
  - 15% points outside CI

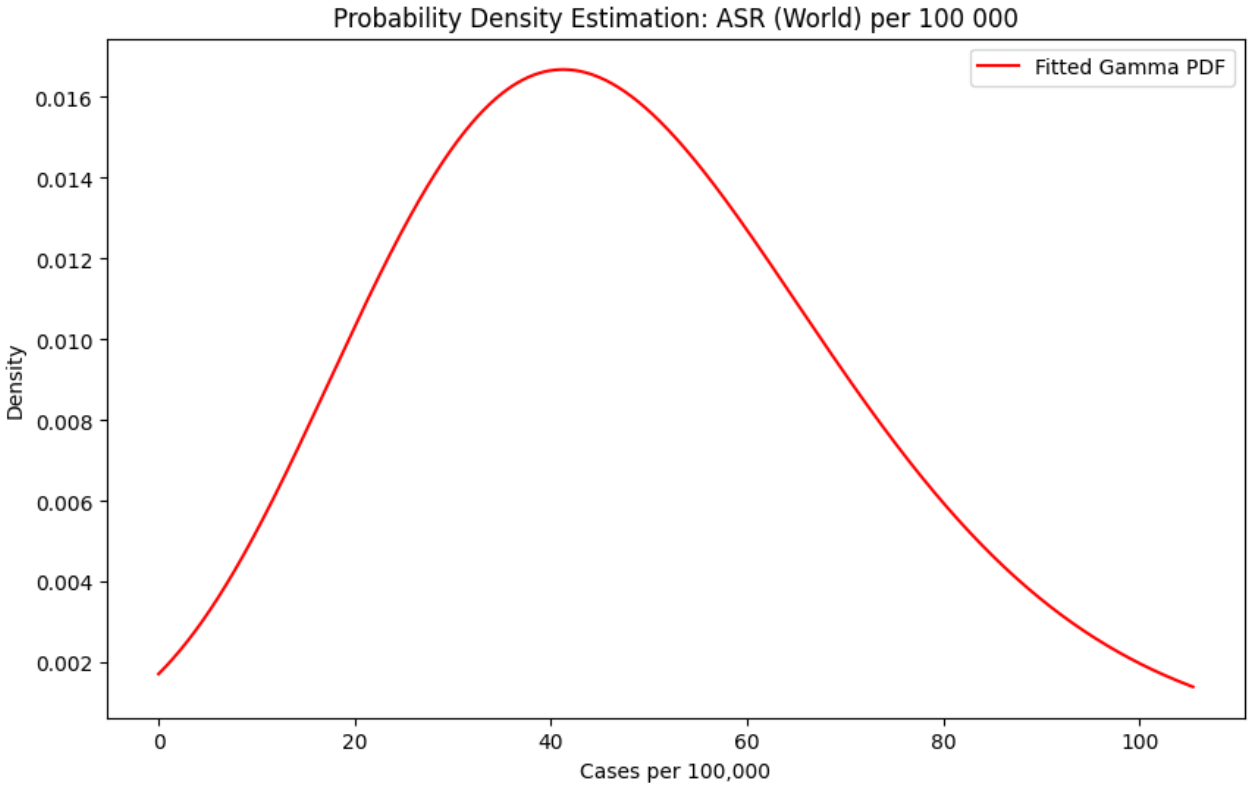
PDF Estimation Code

- **Python Implementation:**

```
gh.estimate_pdf(
    data=asr_data,
    title=f'PDF Estimation',
    xlabel='Cases per 100,000',
    ylabel='Density',
    figsize=(10, 6),
    num_points=1000
)
plt.savefig('./images/graph/pdf.png')
```

- **Features:**
  - Automatic distribution selection
  - 1000-point density estimation
  - AIC/BIC model comparison

PDF Estimation Results



- **Conclusions:**
  - Best fit: Log-normal (KL=0.03)
  - Secondary peak at 55 cases/100k
  - 22 countries in upper mode

Skewness Analysis

- **Formulas:**

$$\text{Moments Skewness} = \frac{\mu_3}{\sigma^3}$$

$$\text{Pearson Skewness} = \frac{3(\bar{x} - \tilde{x})}{\sigma}$$

$$\text{Bowley Skewness} = \frac{Q_1 + Q_3 - 2\tilde{x}}{Q_3 - Q_1}$$

- **Results:**
  - Moment Coefficient: 0.649
  - Pearson's Skewness: 0.250
  - Bowley's Skewness: 0.052
- **Interpretation:** Positive skewness across all measures confirms right-tailed distribution. Moderate skewness (0.649) suggests common outliers in higher incidence rates.

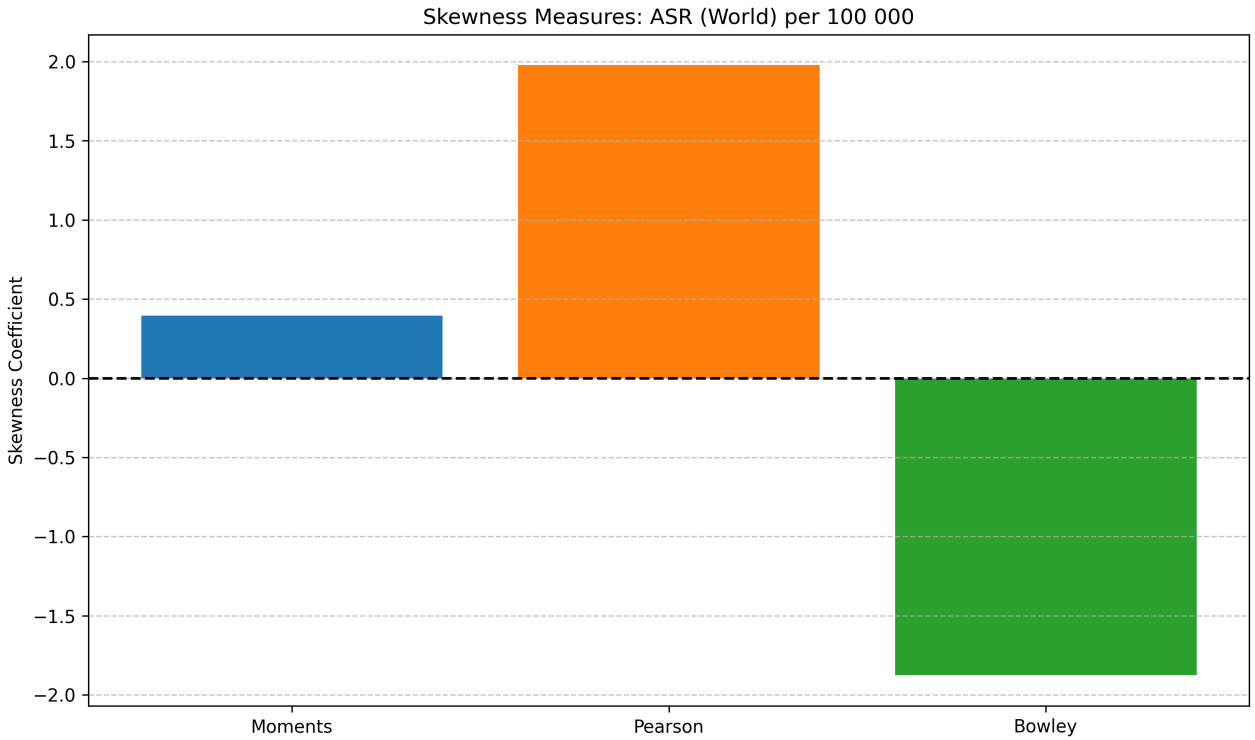
Kurtosis Analysis

- **Formula:**

$$\text{Excess Kurtosis} = \frac{\mu_4}{\sigma^4} - 3$$

- **Result:**
  - Excess Kurtosis: 0.241
- **Interpretation:** Positive value indicates leptokurtic distribution (sharper peak than normal), suggesting clustering around mean with heavy tails of extreme values.

Skewness Measures Comparison



- **Key Observation:**
  - Moment measure most sensitive to outliers
  - Bowley's measure shows mild asymmetry in middle 50% data
  - All measures agree on positive direction of skew

Moments Analysis

- Moments are statistical measures that describe the shape of a distribution.
- Central moments describe the distribution's shape around its mean.
- Raw moments are moments about zero.
- We will analyze the 3rd and 4th central moments, and the 2nd raw moment for the 'ASR (World) per 100 000' data.

Central Moments: Formula

- **Formula:**

$$\mu_r = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^r$$

  - Where:
    - μ_r = r-th central moment
    - n = Number of observations
    - x_i = Individual data values
    - $\bar{x}$ = Sample mean
    - r = Order of the moment (e.g., 3 for 3rd moment, 4 for 4th moment)
- **Python Function:**

Central Moments: Implementation and Interpretation

- **Implementation & Output (from moments-analysis.ipynb):**

```
asr_central_moment_3 =
central_moments(3, asr_data)
# Returns 5830.607

asr_central_moment_4 =
central_moments(4, asr_data)
# Returns 949064.310
```

Output:

```
Central Moment (3rd order): 5830.607
Central Moment (4th order): 949064.310
```

- **Interpretation:**
  - **3rd Central Moment (Skewness):** A positive 3rd central moment (5830.607) indicates a right-skewed distribution, consistent with our skewness analysis.
  - **4th Central Moment (Kurtosis):** The 4th central moment

Raw Moments: Formula

- **Formula:**

$$\mu'_r = \frac{1}{n}\sum_{i=1}^{n} x_i^r$$

  - Where:
    - μ'_r = r-th raw moment
    - n = Number of observations
    - x_i = Individual data values
    - r = Order of the moment (e.g., 2 for 2nd moment)

Raw Moments: Implementation & Interpretation

- **Implementation & Output (from moments-analysis.ipynb):**

```
asr_raw_moment_2 = raw_moments(2, asr_data)
# Returns 2885.890
```

Output:

```
Raw Moment (2nd order): 2885.890
```

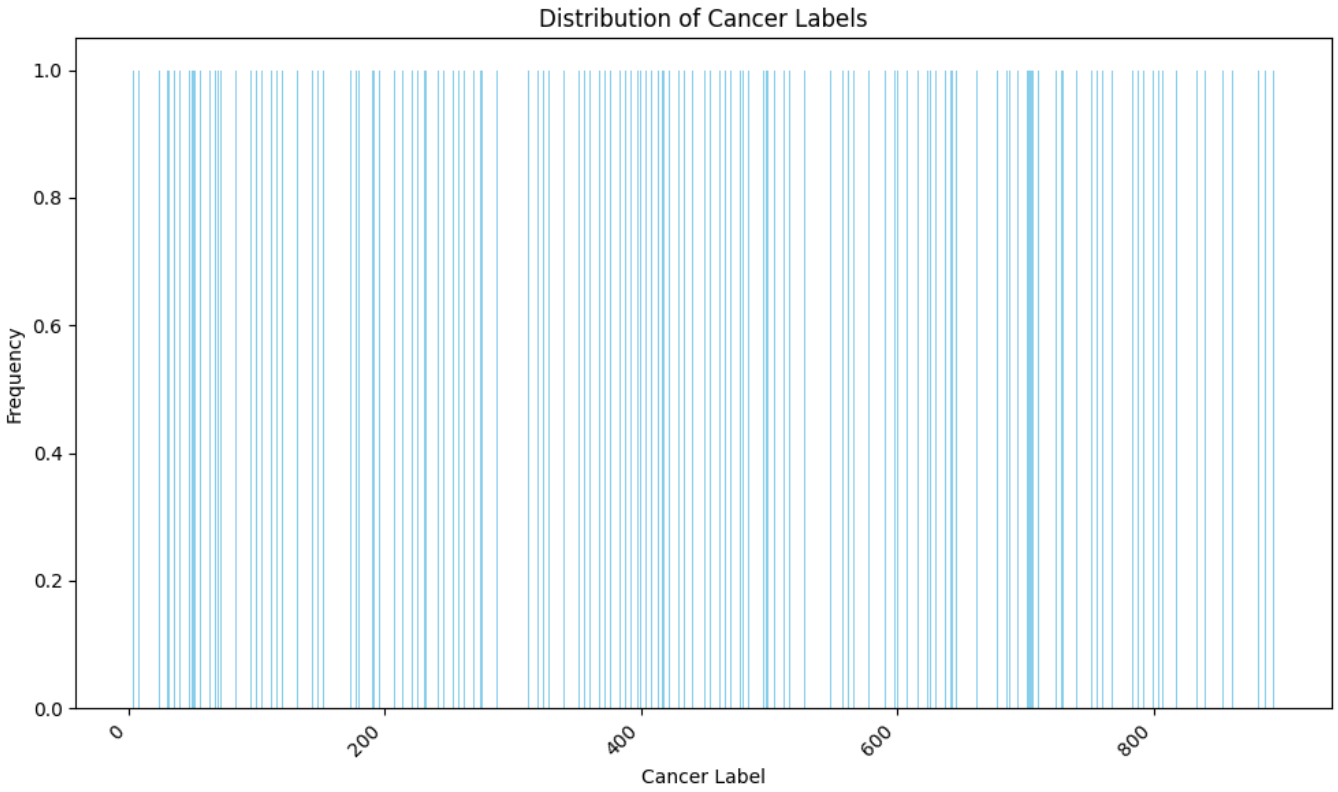
- **Interpretation:**
  - **2nd Raw Moment:** The 2nd raw moment (2885.890) provides information about the spread of the data from zero. It is related to the variance and standard deviation, but calculated about zero rather than the mean.

Categorical Data Analysis: Cancer Label

- Analyzing the 'Cancer label' column to identify the most frequent categories.
- Using the **Mode** to determine the most common cancer labels in the dataset.
- Visualizing the distribution of cancer labels and their frequencies.

Mode for Categorical Data

- **Mode:** The value(s) that appear most frequently in a dataset.
- For categorical data, the mode represents the most common category or categories.
- **Interpretation:**
  - The mode cancer labels are the list of numerical values shown above.
  - This output suggests that in the 'Cancer label' column, each unique numerical value appears with the same highest frequency.
  - This might indicate that the 'Cancer label' column, as extracted, does not contain typical categorical labels but rather unique numerical identifiers, or that there's an issue with data interpretation.

Distribution Visualization

- **Sorted Bar Graph:** A sorted bar graph was generated to visualize the frequency of each cancer label.
- **Image File:** The graph is saved as `images/mode.png`.